



HK IMO Training 2023-2024 Problem Set 10 Suggested Solutions



Problem 37 (Mediterranean MO 2019 Problem 4). Let P be a point inside an equilateral triangle with height 1. Let x, y, z be the distances from P to the three sides of the triangle. Prove that

$$x^2 + y^2 + z^2 \geq x^3 + y^3 + z^3 + 6xyz.$$

Solution. Let $2a$ be the side length of the equilateral triangle, and let A, B, C be the vertices. By considering the areas, we have

$$ax + ay + az = [PBC] + [PCA] + [PAB] = [ABC] = a.$$

This implies $x + y + z = 1$. Therefore,

$$\begin{aligned} x^2 + y^2 + z^2 &= (x^2 + y^2 + z^2)(x + y + z) \\ &= x^3 + y^3 + z^3 + (x^2y + x^2z + y^2x + y^2z + z^2x + z^2y) \\ &\geq x^3 + y^3 + z^3 + 6xyz \end{aligned}$$

by the AM-GM inequality. Equality holds when $x = y = z = \frac{1}{3}$. □

Problem 38 (Czech-Polish Slovak Junior Match 2019 Team Problem 1). Let x and y be rational numbers such that both $x + y$ and $x^2 + y^2$ are integers. Prove that x and y are integers.

Solution. Let $x = \frac{a}{b}$ and $y = \frac{c}{d}$ where $(a, b) = (c, d) = 1$ and $b, d > 0$. Firstly, as $x^2 + y^2 = \frac{a^2d^2 + b^2c^2}{b^2d^2}$ is an integer, we know that $b^2 \mid a^2d^2 + b^2c^2$, and hence $b^2 \mid a^2d^2$. As $(a, b) = 1$, we must have $b^2 \mid d^2$. Thus, $b \mid d$. By symmetry, we have $d \mid b$ as well. Therefore, $d = b$.

Secondly, note that $(x + y)^2 - (x^2 + y^2) = 2xy = \frac{2ac}{b^2}$ is an integer. This implies $b^2 \mid 2ac$. As $(a, b) = (c, b) = 1$, this yields $b^2 \mid 2$, and hence $b = 1$. This means both x and y are integers. □

Problem 39 (Serbia MO 2015 Problem 3). There are 100 prisoners. Each prisoner is given a hat coloured in one of five colours. Everyone can see all hats except his own. Afterwards, the guard asks the prisoners one by one whether he knows the colour of his hat or not. If the prisoner answers 'no', he will be put in the jail. If the prisoner answers 'yes', he needs to write down the colour that he guesses. If this colour agrees with the colour of his hat, he will be freed, otherwise he will be put in the jail. Suppose all other prisoners can hear the answer of the prisoner but cannot see the colour he writes down, and they can know the outcome immediately. If the prisoners can think of a strategy together before they are given the hats, but cannot communicate afterwards, what is the maximum number of prisoners who are guaranteed to be freed?

Solution. The answer is 98.

Let P_j be the j th prisoner. Clearly, there is no way to guarantee P_1 to be freed. Since there are only 3 outcomes of the first prisoner (answering 'no', answering 'yes' and being put in the jail, or answering 'yes' and being freed) but there are 5 possible colours, we cannot guarantee P_2 to be freed as well. Therefore, at most 98 prisoners are guaranteed to be freed.

It remains to provide a strategy that can always free all prisoners except P_1 and P_2 . Firstly, we use 0, 1, 2, 3, 4 to denote the colours of the hats. Let x be the colour of P_2 's hat. Let y be the sum of colours of the hats of $P_3, P_4, \dots, P_{100}$ modulo 5. We provide a strategy such that everyone will know the value of y after P_1 and P_2 have given their answers. If this can be done, then the remaining prisoners can work out the colours of their hats. (For example, P_3 knows the colours of hats of $P_4, P_5, \dots, P_{100}$. The value of y then uniquely determines the colour of P_3 's hat.)

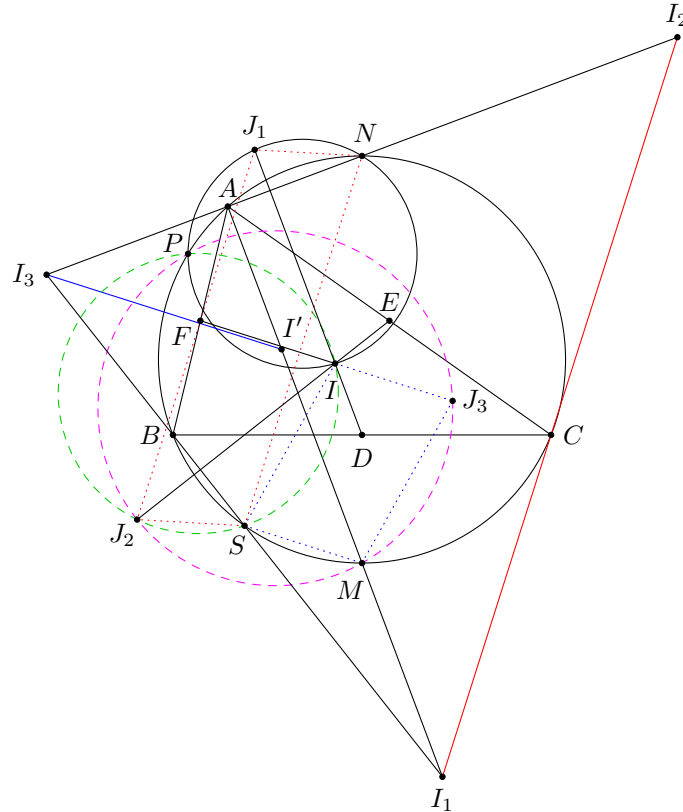
We provide the answers of P_1 and P_2 in the table below. (Notations: N denotes 'no', Y denotes 'yes', and the number indicates the colour that P_2 guesses after answering 'yes'.) Note that P_1 knows the values of x and y , P_2 only knows the value of y (hence his answer in each column can only depend on P_1 's answer), and all other prisoners know the value of x (and they do not know the colour guessed by P_2 if P_2 answers 'yes').

	$y = 0$	$y = 1$	$y = 2$	$y = 3$	$y = 4$
$x = 0$	N, N	N, Y0	Y, N	Y, Y1	N, Y2
$x = 1$	N, N	Y, Y3	Y, N	Y, Y1	N, Y2
$x = 2$	Y, N	Y, Y3	N, N	N, Y3	N, Y2
$x = 3$	Y, N	Y, Y3	N, N	N, Y3	Y, Y4
$x = 4$	Y, N	N, Y0	N, N	Y, Y1	Y, Y4

- If P_2 answers ‘no’, the only possible values of y are 0, 2. Based on P_1 ’s answer, everyone can deduce the exact value of y .
- If P_2 answers ‘yes’ and is freed, the only possibilities are $(x, y) = (0, 1), (1, 3), (2, 4), (3, 1), (3, 3), (4, 4)$. If $x = 0, 1, 2, 4$ (which is known to the rest), everyone knows the value of y . If $x = 3$, everyone can deduce the exact value of y based on P_1 ’s answer.
- If P_2 answers ‘yes’ and is put in the jail, the only possibilities are $(x, y) = (0, 3), (0, 4), (1, 1), (1, 4), (2, 1), (2, 3), (3, 4), (4, 1), (4, 3)$. For each choice of x , there are at most two possibilities. One can always deduce the exact value of y based on P_1 ’s answer.

Therefore, at least 98 prisoners are guaranteed to be freed under this strategy. $\square$

Problem 40 (Malaysia TST 2023 Problem 3). Let $\triangle ABC$ be an acute triangle with $AB \neq AC$. Let D, E, F be the midpoints of BC, CA, AB respectively. Let M and N be the midpoints of the arc BC of the circumcircle of $\triangle ABC$ such that M and A lie on different sides of BC , while N and A lie on the same side of BC . Let I, J_1, J_2, J_3 be the incentre, D -excentre, E -excentre and F -excentre of $\triangle DEF$ respectively. Prove that the circumcircles of $\triangle ABC$, $\triangle NIJ_1$ and $\triangle MJ_2J_3$ meet at a common point.



Solution. Let I', I_1, I_2, I_3 be the incentre, A -excentre, B -excentre and C -excentre of $\triangle ABC$ respectively. Let S be the midpoint of the arc ABC of (ABC) . Let P be the second intersection point of (ABC) and (NIJ_1) . We need to prove P, J_2, M, J_3 are concyclic.

We first prove that J_1J_2SN is a parallelogram. There is an obvious homothety between $\triangle DEF$ and $\triangle ABC$ with ratio -2 . It maps J_1J_2 to I_2I_1 . This shows $J_1J_2 \parallel I_2I_1$ and $2J_1J_2 = I_2I_1$. It is well-known that S and N are the midpoints of I_1I_3 and I_2I_3 respectively. Therefore, $NS \parallel I_2I_1$ and $2NS = I_2I_1$. Combining these, we see that J_1J_2 and NS have the same direction and length. Thus, J_1J_2SN is a parallelogram.

Next, we prove that $ISMJ_3$ is a parallelogram. Again, by homothety, we have $IJ_3 \parallel I_3I'$ and $2IJ_3 = I_3I'$. Also, S and M are the midpoints of I_1I_3 and I_1I' respectively. Therefore, $SM \parallel I_3I'$ and $2SM = I_3I'$. Again, this shows IJ_3 and SM have the same direction and length, and hence $ISMJ_3$ is a parallelogram.

The rest is angle chasing. Using directed angles, as I is the incentre of $\triangle DEF$, we have

$$\angle DIE = \frac{\pi}{2} + \angle IFE = \frac{\pi}{2} + \angle(IF, EF) = \frac{\pi}{2} + \angle(SM, BC) = \angle(SM, MN) = \angle SMN. \quad (1)$$

Using this, we find that

$$\begin{aligned} \angle SJ_2I &= \angle(SJ_2, J_2I) \\ &= \angle(NJ_1, J_2I) \\ &= \angle(NJ_1, J_1I) + \angle(J_1I, J_2I) \\ &= \angle NJ_1I + \angle J_1IJ_2 \\ &= \angle NPI + \angle DIE \\ &= \angle NPI + \angle SMN \\ &= \angle NPI + \angle SPN \\ &= \angle(NP, PI) + \angle(SP, PN) \\ &= \angle(SP, PI) \\ &= \angle SPI. \end{aligned}$$

This implies P, I, S, J_2 are concyclic. Again using (1), we deduce

$$\angle SNM = \frac{\pi}{2} + \angle SMN = \frac{\pi}{2} + \angle DIE = \frac{\pi}{2} + \angle J_1IJ_2 = \frac{\pi}{2} + \angle(J_1I, IJ_2) = \angle(J_3J_2, IJ_2) = \angle J_3J_2I.$$

Therefore, we have

$$\begin{aligned} \angle J_2PM &= \angle(J_2P, PM) \\ &= \angle(J_2P, PS) + \angle(PS, PM) \\ &= \angle J_2PS + \angle SPM \\ &= \angle J_2IS + \angle SNM \\ &= \angle J_2IS + \angle J_3J_2I \\ &= \angle(J_2I, IS) + \angle(J_3J_2, J_2I) \\ &= \angle(J_3J_2, IS) \\ &= \angle(J_3J_2, J_3M) \\ &= \angle J_2J_3M. \end{aligned}$$

This implies P, J_2, M, J_3 are concyclic as desired. □